Adaptive Policy Optimization for Municipal Solid Waste Segregation using Coupled Agent-Based Modeling and Deep Reinforcement Learning

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***Abstract*—As municipalities strive to build a resilient and sustainable society, effective municipal solid waste (MSW) segregation emerges as a critical sustainability bottleneck. Applying standard Deep Reinforcement Learning (DRL) to optimize public policy in these multi-agent human environments introduces severe computational challenges, primarily due to the sparse reward problem caused by cultural behavioral inertia. To address this, we introduce a novel socio-technical framework coupling an Agent-Based Model (ABM) with a Heuristic-Guided DRL (HuDRL) agent, functioning as an intelligent system for predictive municipal governance. Using the MSW segregation mandate in Bacolod, Lanao del Norte as a complex test environment, this study optimizes public resource allocation within a decentralized, polycentric governance architecture. Household cognition is formalized via a dynamically weighted Theory of Planned Behavior (TPB), where a Global Sensitivity Analysis mathematically proves that the internal friction of washing, sorting, and storing waste is the dominant driver of non-compliance, accounting for 79% of system variance. To navigate this non-convex policy space, the central Markov Decision Process (MDP) utilizes safety-critical action masking and a targeted amplification algorithm. Comparative simulations reveal that standard DRL algorithms and status quo heuristic policies stagnate in a "Collapse Zone" (defined operationally as sustained global compliance below 25% for multiple epochs). Conversely, the HuDRL agent successfully converges on a "Sequential Saturation" strategy (i.e., concentrating majority funding into single nodes until specific compliance thresholds are breached). By leveraging localized social tipping points, the algorithm proves that sequentially concentrating municipal funds is vastly superior to equitable distribution, directly advancing Sustainable Development Goals 11, 12, and 16 while establishing a predictive paradigm for algorithmic governance.**

***Index Terms*—Agent-Based Modeling, Deep Reinforcement Learning, Intelligent Systems, Municipal Solid Waste Management, Polycentric Governance, Sparse Reward Problem, Sustainable Society, Theory of Planned Behavior.**

I. Introduction

Building a resilient and sustainable society requires intelligent systems capable of navigating and optimizing multi-agent human environments. However, doing so presents a profoundly complex computational challenge. Unlike static physical environments, socio-economic systems are characterized by bounded rationality, delayed feedback loops, and profound cultural inertia. When policymakers attempt to alter deeply ingrained community behaviors to solve sustainability problems, they operate within a non-convex policy space where human intuition—often biased toward equitable, simultaneous resource distribution—consistently leads to suboptimal outcomes.

From a computational perspective, applying standard Deep Reinforcement Learning (DRL) algorithms to these environments introduces a severe manifestation of the "Sparse Reward Problem" [8]. Because human behavioral shifts require sustained, concentrated pressure before triggering a measurable state change, standard AI agents often fail to establish causal links between their short-term budgetary actions and long-term environmental rewards.

To computationally solve this policy implementation deficit, this study utilizes the management of Municipal Solid Waste (MSW) as a high-fidelity sustainability test environment. In the Philippines, the Ecological Solid Waste Management Act of 2000 (R.A. 9003) mandates household-level segregation. However, Local Government Units (LGUs) struggle with systemic non-compliance, demonstrating what we term the "MRF Paradox"—a scenario where mandated Material Recovery Facilities (MRFs) increased nationwide by 8.7% between 2023 and 2024, yet illegal dumpsites concurrently surged by 84%. Frontline municipal bodies are trapped in a system of trial-and-error budgeting. Furthermore, the advent of the E-Governance Act of 2026 has fundamentally shifted the administrative landscape. While national law mandates digitalization, the operational burden of accountability and liability for systemic policy failure remains a critical gap at the local level. Trial-and-error resource allocation is no longer legally or fiscally viable; LGUs require advanced predictive analytics to operationalize accountability and mathematically justify public expenditure.

We propose a coupled Agent-Based Modeling (ABM) and Deep Reinforcement Learning (DRL) architecture. By integrating the Theory of Planned Behavior (TPB) into the ABM, we create a virtual laboratory that accurately simulates structural friction and social hysteresis. Operating atop this environment, a Heuristic-Guided DRL (HuDRL) agent representing the central municipal authority dynamically optimizes continuous budget allocations. The primary objective of this study is to engineer an intelligent algorithmic governance framework capable of systematically overcoming behavioral inertia without breaching the fiscal realities of local municipalities.

II. Related Work

A. Behavioral Friction and Social Tipping Points

Designing cost-efficient policies requires models grounded in psychological theory. The Theory of Planned Behavior (TPB) posits that an individual's intention to perform a behavior is a function of Attitude, Subjective Norms, and Perceived Behavioral Control [7]. However, translating intention into action is frequently obstructed by logistical realities. In environmental compliance, positive intentions are easily overridden by the "Cost of Effort" [12].

Furthermore, sociologists recognize that sustained behavioral change relies on critical mass. Once a committed minority reaches a social tipping point—historically approximated at 25% to 40%—it triggers a rapid cascade of adoption across the majority [3]. This transitions behaviors from calculated decisions into ingrained cultural norms, generating a "Social Norm Shield" (defined in this study as a behavioral lock-in state where local compliance $\geq 70\%$, heavily resisting negative behavioral decay) [4].

B. Algorithmic Policy Design

Traditional optimization methods, such as Linear Programming, assume perfect rationality and fail to capture the non-linear hysteresis of human society [2]. Agent-Based Models (ABM) have emerged as powerful virtual laboratories for socio-environmental systems, but standard implementations require manual scenario testing.

Recent advancements demonstrate the efficacy of formalizing multi-agent environments as a Markov Decision Process (MDP), allowing neural networks to autonomously discover adaptive strategies in high-dimensional spaces [8]. However, standard Proximal Policy Optimization (PPO) algorithms deployed in human policy contexts require heuristics and safety-critical action masking to prevent convergence failure and ensure rigid fiscal boundaries are respected during training.

III. System Architecture and The Simulation Gymnasium

This study integrates ABM with DRL optimization within a strict polycentric governance framework. The simulation is geographically bounded to Bacolod, Lanao del Norte, a 4th class municipality. It operates on a spatially explicit, non-toroidal 50×50 grid containing 4,022 Household Agents partitioned into seven distinct component barangays (Poblacion, Liangan East, Ezperanza, Binuni, Demologan, Mati, and Babalaya).

The architecture utilizes a dual-clock temporal resolution to bridge micro-behavior and macro-policy. Household agents execute their decision-making logic on daily micro-ticks, simulating the continuous generation and management of waste. Conversely, the central Mayor Agent evaluates the environment and executes budget distributions on quarterly (90-day) epochs over a 1,080-tick (3-year) simulation horizon.

The core architectural novelty is the continuous feedback loop between the discrete ABM and the continuous MDP. To prevent the neural network from being overwhelmed by the micro-state data of 4,022 households, the architecture utilizes meso-level Barangay Agents as data aggregators. These agents compress the daily micro-behavioral outputs into a unified 17-dimensional state vector in $\mathcal{O}(N)$ time, providing a clear observation space for the Mayor Agent's MDP. The complete polycentric simulation architecture is illustrated in Figure 1.

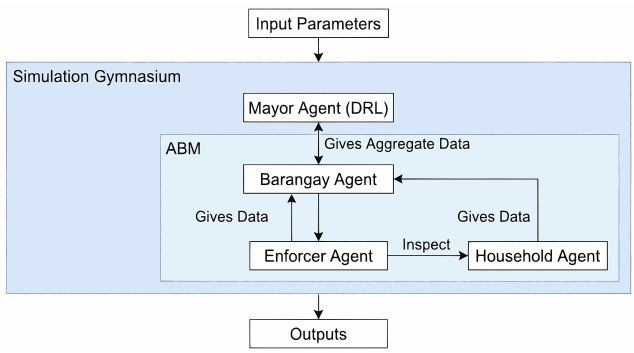

Fig. 1. Simulation Architecture Diagram

IV. The Multi-Agent Environment

A. Household Agents and the Theory of Planned Behavior

Household agents operate as continuous finite state machines. To ensure population heterogeneity, behavioral weights are instantiated via stochastic Gaussian sampling $\mathcal{N}(\mu, \sigma^2)$, bounded within $[0.1, 0.6]$. Their cognition is governed by a dynamically weighted TPB utility equation (U_{seg}):

$$U_{seg} = (w_A A) + (w_{SN} SN_{loc}) + (w_{PBC} PBC) - C_{Net} + \epsilon \quad (1)$$

where A, SN_{loc}, and PBC represent Attitude, localized Subjective Norms, and Perceived Behavioral Control, buffered by stochastic human noise (ϵ).

To allow the DRL agent to interact with psychological barriers, we mathematically formalized the Net Cost (C_{Net}) function:

$$C_{Net} = C_{Effort} + \gamma(C_{Mon} - I - F \cdot P_{Detect}) \quad (2)$$

Crucially, C_{Effort} is strictly defined not as external travel distance, but as the internal household friction of washing, sorting, and storing segregated waste. This physical hassle is weighed against financial variables: tangible costs (C_{Mon}), municipal incentives (I), and expected fines ($F \times P_{Detect}$) [6]. These financial vectors are scaled by γ, an income-sensitivity multiplier.

To model cultural inertia, agents query their spatial environment using an $\mathcal{O}(k)$ Moore neighborhood algorithm. The simulation incorporates a "Bayanihan" transitional state, establishing a 40% compliance floor. If local compliance breaches the critical 70% tipping point ($SN_{loc} \geq 0.70$), the behavior internalizes, mathematically hard-locking the subjective norm weight ($w_{sn} = 0.90$). The complete daily execution is detailed in Algorithm 1.

Algorithm 1 The Household's Daily Decision Process

Input: Household attitude, neighborhood compliance, and internal friction
Step 1: Assess Peer Pressure
Scan local neighborhood to determine compliance rate
if local compliance $\geq 70\%$ **then**
  Internalize habit: Lock peer pressure influence to maximum
else if local compliance $< 40\%$ **then**
  Shield failure: Rely strictly on the fear of enforcement
end if
Step 2: Weigh Costs and Intentions
Calculate total motivation (attitude plus peer pressure)
Calculate net cost (internal friction plus expected fines, minus incentives)
Step 3: Execute Action
if total motivation outweighs the net cost **then**
  Action: Segregate waste (accounting for rare human error)
else
  Action: Remain non-compliant
end if

B. Enforcement and Barangay Agents

Enforcement agents physicalize the P_{Detect} risk profile. Operating under bounded rationality, they execute stochastic random walks and localized spatial queries (Moore neighborhood radius $r = 10$) rather than possessing $\mathcal{O}(1)$ global oversight. The model differentiates between *Micro-Enforcement* (local Tanods with a 90% distraction rate) and *Macro-Enforcement* (LGU Strike Force with a cost of ₱12,000 per enforcer and ₱1,000 per citation). Their deployments are bound by strict 30-day temporal contracts to simulate real-world municipal payroll realities [11].

Barangay Agents serve as meso-level administrative nodes executing a polycentric Cascading Incentive Distribution Protocol. When households claim rewards, the agent draws first from its static Internal Revenue Allotment (IRA) and second from the Mayor's dynamic augmentation budget. If both are exhausted, the household suffers a computational "Perceived Unfairness" penalty, severely degrading intrinsic motivation [12].

V. The Deep Reinforcement Learning Agent

A. MDP Formulation

The central policymaker is formalized as an episodic MDP defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$:

- **State Space ($\mathcal{S}$):** A 17-dimensional continuous vector representing the aggregated quarterly compliance of all 7 barangays, localized budget exhaustion rates, time step, and Political Capital.
- **Action Space ($\mathcal{A}$):** A 21-dimensional continuous vector ($At \in [-1, 1]^{21}$) indicating budget allocations across IEC, Enforcement, and Incentives for all 7 barangays.

The central constraint is an annual municipal budget of ₱1,500,000, discretized to a maximum quarterly operating budget (B_q) of ₱375,000. The PPO Actor network's output is processed through a Temperature-Scaled Softmax function ($\tau = 5.0$) to guarantee the multi-dimensional action vector sums to exactly 100% of available funds, bounding the search space and strictly enforcing the fiscal reality.

B. Multi-Objective Reward Shaping and Heuristics

To solve the sparse reward problem inherent in directing human cultural shifts, the MDP utilizes a composite reward function (R_t):

$$R_t = R_{Base} + R_{Jackpot} + R_{Focus} - P_{Waste} + \Phi_{cap} \quad (3)$$

Scalar weights balance environmental and fiscal mandates. Crucially, Φ_{cap} serves as a Political Capital Modifier; if public approval drops below 10% due to excessive fine issuance, it forces an immediate administrative collapse penalty.

To prevent the AI from defaulting to equitable distribution, the reward structure incorporates heuristic safety guardrails. As detailed in Algorithm 2, the 70% Graduation Mechanism slashes budget allocations to saturated barangays by 99%, artificially forcing the neural network to pivot its fiscal arsenal to the next bottleneck.

Algorithm 2 The Mayor Agent's Triage & Safety Heuristics

Input: Current compliance levels and AI's proposed budget
Rule 1: The Graduation Mechanism
for each barangay **do**
  if barangay compliance $\geq 70\%$ **then**
    Slash its proposed allocation by 99% to force AI to pivot funds
  end if
end for
Rule 2: The Phase-Shift Crisis Guardrail
if global municipal compliance $< 40\%$ **then**
  Identify the weakest performing barangay
  Guarantee it receives at least half of the total budget
end if
Rule 3: Strict Fiscal Normalization
Proportionally scale all allocations so they sum to exactly 100%
return Final executed budget distribution

VI. Results and Discussion

A. Calibration and Global Sensitivity Analysis

Empirical calibration anchored the model utilizing Municipal Environment and Natural Resources Office (MENRO) audits, establishing a valid Status Quo baseline of 12.5% global compliance (counteracting inflated self-reports of over 50%) [5].

A Sobol Global Sensitivity Analysis (GSA), summarized in Table I, revealed that the internal "Cost of Effort" (C_{effort}) is the overwhelmingly dominant driver of behavior. Yielding a Total-Order index (S_T) of 1.06 and accounting for 79% of the variance in global compliance ($S_1 \approx 0.79$), this mathematically validates the Intention-Action gap: if the

internal friction of washing and sorting is too high, households default to non-compliance regardless of environmental beliefs. Conversely, intrinsic Attitude weight (w_A) exhibited near-zero sensitivity, proving that Information Campaigns (IEC) suffer strict diminishing returns without localized structural support.

TABLE I
SOBOL GLOBAL SENSITIVITY INDICES ($N = 10,000$ SAMPLES)

Input Parameter	**First-Order** (S_1)	**Total-Order** (S_T)
Attitude Weight (w_A)	-0.01	0.10
Subj. Norm Weight (w_{SN})	0.06	0.21
PBC Weight (w_{PBC})	-0.01	0.01
Cost of Effort (C_{Effort})	**0.79**	**1.06**
Decay Rate (λ)	0.01	0.00

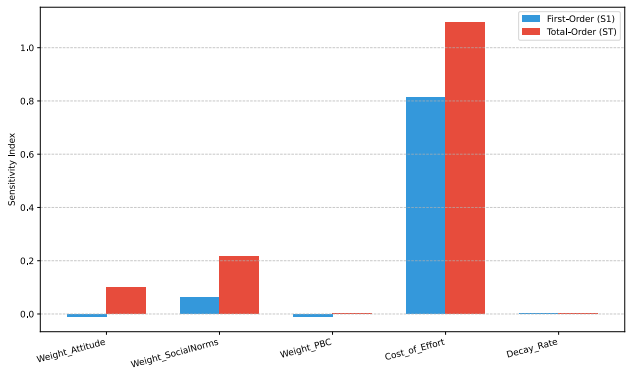

Fig. 2. General Sensitivity Analysis using the Sobol Method

B. Computational Convergence

The simulation benchmarked the HuDRL architecture against a baseline Vanilla PPO approach over a 12-quarter horizon (Figure 3). The baseline PPO exhibited severe policy stagnation, trapped in the "Collapse Zone" (operationally defined as sustained global compliance remaining below 25% for ≥ 3 consecutive quarters) for the entire 3-year timeline. Without action-masking, the standard AI failed to concentrate enough capital to break cultural inertia.

In stark contrast, the HuDRL agent demonstrated rapid gradient convergence. Bounded by the Graduation Mechanism, the network efficiently navigated the non-convex space, breaching the 70% Stability Threshold by Quarter 2 and reaching asymptotic maintenance saturation (98%) by Quarter 6.

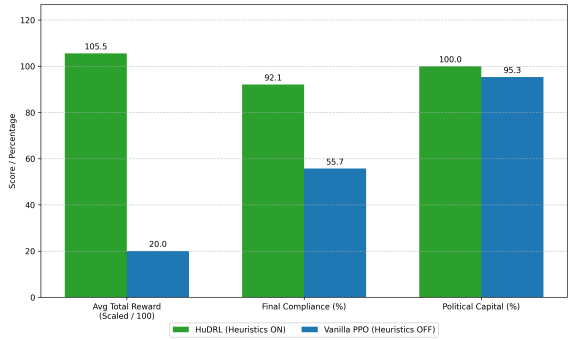

Fig. 3. Performance Comparison of HuDRL vs. Baseline PPO

C. Comparative Analysis of Policy Strategies

Simulation experiments over a three-year horizon demonstrated a stark divergence between intuitive heuristics and DRL optimization (Figure 4).

The *Status Quo* baseline policy, distributing funds equally across barangays (₱53,571.43 per locale), acted as a mathematical dilution mechanism. It created an "Urban Resource Trap" (defined here as a state where equitable budget distribution dilutes per-capita impact below the minimal threshold required to shift behavioral hysteresis) in dense areas like Poblacion, failing entirely to cross the behavioral tipping point and terminating at 30.18% global compliance. A *Pure Enforcement* strategy triggered the "Draconian Trap" (where aggressive penalization spikes short-term compliance but severely degrades civic trust), draining public approval to cause a "Political Collapse" (an absorbing simulation failure state triggered when public approval $\leq 10\%$) by Quarter 2. Conversely, a *Pure Incentives* strategy triggered the "Transactional Trap" (where immediate budget exhaustion on transient rewards fails to internalize habits), draining budgets instantly and stalling compliance at 15.91%.

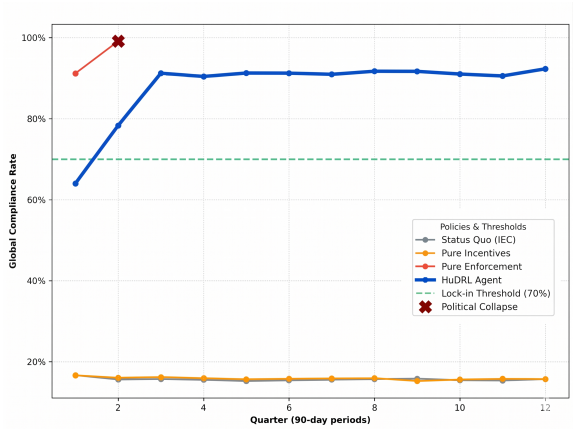

Fig. 4. Compliance Rate Performances of Simulated Policy Strategies

The HuDRL Mayor Agent autonomously discovered an aggressive *"Sequential Saturation"* strategy (formally defined as allocating $\geq 60\%$ of the quarterly budget to a single underperforming node until its local compliance breaches 70%), achieving 90.45% sustained compliance. Bypassing human bias toward simultaneous fairness, the AI sequentially concentrated up to 69% of its budget onto single failing urban nodes.

Once a locale breached the 70% norm internalization threshold, the AI ceased active funding—exploiting the Social Norm Shield (the mathematical lock on subjective norm weights triggered at $\geq 70\%$ compliance) to prevent decay—and pivoted its fiscal arsenal to besiege the next municipal bottleneck. This transition from trial-and-error budgeting to Algorithmic Resource Targeting provides the necessary predictive layer for LGUs to operationalize accountability.

VII. Conclusion

This study validates that coupling Agent-Based Modeling with Heuristic-Guided DRL provides a mathematically rigorous virtual laboratory for complex public policy optimization. By formalizing municipal dynamics into an MDP, the framework demonstrated that standard DRL algorithms fail in multi-agent socio-technical systems due to sparse rewards and cultural hysteresis. The optimized HuDRL architecture proved that overcoming the internal friction of household segregation requires targeted, sequential saturation of resources, which is vastly superior to broad, diluted mandates. By deploying this framework as an intelligent system for resilient municipal governance, LGUs can transition from reactive trial-and-error budgeting to precision-targeted mathematical strategies. Ultimately, this approach directly aligns with the imperative to build a resilient and sustainable society, paving the way for future research to explore transitioning this architecture into a full Municipal Digital Twin by integrating Vehicle Routing Problems (VRP) and expanding into Multi-Agent Reinforcement Learning (MARL) for commercial actors.

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USE OF GENERATIVE AI

The authors used Gemini Pro solely for language refinement and did not contribute to the intellectual content or analysis of the paper.

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